

A system dynamics approach to fast fashion

Abstract:

The fast-fashion industry has become increasingly relevant in the past five years, providing rapid, low-cost clothing production to various parts of the world with significant environmental consequences. The industry contributes to detrimental levels of water consumption and pollution, GHG emissions, and textile waste generation, alongside unethical social and labour practices. The objective of the study is to model the dynamic interactions between consumer behaviour, production processes, and their environmental impacts to identify key leverage points within the system for future policies, alongside an evaluation of the current sustainability interventions employed by various fast-fashion brands to determine their effectiveness in mitigating the negative consequences of the industry. For this, a systems dynamic approach is modeled via Vensim to simulate the complex interdependencies that include consumer adoption, resource consumption, environmental impacts (as mentioned above), and sustainability practices. The model links the bass diffusion process, item consumption rate, and the corresponding impact using existing data on consumer patterns integrated into the model to show their behaviour. The results highlight the significant impact of consumer adoption dynamics on the overall environmental load within the next 4 years and identify current policies as inefficient in decreasing waste generation. Furthermore, the model suggests that policies focusing on extending average item lifespan and enhancing consumer awareness are particularly effective in decreasing the average GHG emissions. For this particular model, the average consumption rate per adopter and adoption from word of mouth are considered to be critical drivers of the system's behaviour.

Introduction:

The fast fashion industry - mainly comprising brands like SHEIN, Temu, Zara, and others - focuses on getting new styles to consumers at a record pace. It consists of ultralow prices and condensed production cycles at the cost of the environment. The industry contributes towards 10% of the global carbon emissions, dries up water sources, pollutes river streams, and releases 500,000 tons of microfiber into the ocean each year. Furthermore, it has the largest impact on freshwater withdrawal and depletion of ecosystem quality, while also being responsible for the highest impact on resource depletion due to the energy-intensive processes based on fossil fuel energy. The industry also poses societal problems, promoting child labour in third-world countries, conducting operations in poorly developed infrastructure, and rarely following environmental regulations due to relaxed laws and regulations, and profits carrying more precedent than human welfare. The purpose of this study focuses on the environmental degradation that is a result of the current demand cycle and the policies that affect it. Particularly, the general impact of the fast fashion market on GHG emissions, waste generation, water usage, and water pollution. Later on, these impacts are evaluated by introducing policy measures taken by the mentioned brands and simulating the extent to which they are useful. The current scope of the model uses the Bass

Diffusion process to model demand surges as the demand cycles change frequently. This is further modified by including various trend cycles and seasonality to factor in the nature of the market. Although each particular environmental consequence poses an issue on its own, a model explaining and evaluating the impact of policy interventions and creating a developed understanding of the consumer's role is only efficiently applicable when multiple environmental consequences are modeled alongside it for a good overview. Therefore, the social impact in terms of production and supply chain is not taken into consideration in the current scope of the model. In particular, the broad overview of the model is used for the evaluation of current strategies alongside future strategies that will remain pivotal to mitigate the environmental impact of the industry. A holistic overview of the model can be explained with the problem statement described below:

"How do the interconnected systemic dynamics within the fast fashion industry, driven by consumer behavior and influenced by existing policy interventions, contribute to environmental degradation and what alternative, holistic systems-based interventions can be identified and evaluated to more effectively mitigate these impacts?"

Demographics:

The current model used is integrated with secondary data for the European Union taking into consideration that a majority of the fast fashion market is present there. It is assumed that a base income level for all consumers remains at the median value of 1500 euros where all consumers are able to afford low-cost fast fashion goods at an average of 3-4 purchases a month. The age distribution considered is between 18-35 years of the EU population with all consumers assumed to have an equal level of awareness of fast fashion brands. The current data cited is benchmarked via the Inditex corporation that acts as the umbrella company for a majority of the industry.

Literature Review:

A systematic review by Bailey et al., which reportedly synthesized approximately 3,315 articles down to 65 key case studies, aimed to model the global environmental impact of fast fashion, with a particular focus on water resources (Bailey et al.). Their research quantifies significant environmental burdens, providing crucial data for our model's stocks and variables. Key figures attributed to this synthesis include fast fashion's contribution of 8% to 10% of total global Greenhouse Gas (GHG) emissions, annual water consumption of 93 billion cubic meters (m³), and the generation of 92 million tons of textile waste (Bailey et al.). While this data aggregates findings from multiple studies, our research adapts these global figures to a system dynamics framework specifically for the European Union (EU). Furthermore, Bailey et al. outline fundamental policy interventions, such as enhancing consumer awareness, promoting the '3 Rs' (Reduce, Reuse, Recycle), and encouraging the extension of product lifespans, supported by programs like the Waste and Resources Action Programme (WRAP) (Bailey et al.).

Building on the need for systemic understanding, Valterre et al. developed a comprehensive system dynamics model of the EU textile system (Valterre et al.). Although their model does not feature explicitly labeled equations, it clearly defines stocks—such as "clothing in use" and "landfilled clothing"—and the flows that affect them, like the "item discard rate" (draining

clothing in use) and the "landfill flow." A primary objective of Valtere et al.'s study was to quantify the effects of interventions, notably reduced clothing consumption, on the annual volume of landfill waste within the EU. Their analysis involved three scenarios: a baseline with no policy intervention; a scenario introducing a sorting strategy; and a third where reduced clothing consumption was triggered by an awareness campaign, itself activated when landfill accumulation reached a predefined threshold. The findings from Valtere et al. are particularly compelling, suggesting that policies focused on consumption reduction could lower annual landfill waste in the EU to 0.7 million tonnes by 2030, significantly outperforming strategies based solely on waste sorting (Valtere et al.). Our model attempts to combine all three scenarios with the addition of other intervention strategies to outline the combined effect on the emission, landfill, and water stocks. These strategies include brand transparency, circularity promotion initiatives, and sustainable product options introduced to the consumer.

Further quantitative insights for our model are drawn from a report by the Centre for the Promotion of Imports from developing countries (CBI). This report offers data on fast fashion trends and the impact of awareness campaigns. For instance, one cited effect of an awareness campaign was an 8% increase in the uptake of sustainable products, which we translated into an "awareness" variable of 0.005 within our model (CBI). The CBI report also discusses the quantified impacts related to transparency and circularity in the fashion industry. Complementing this, a report by Martina Igini (associated with Earth.Org) provides critical statistics on textile waste, microplastic pollution, average item lifespan, and GHG emissions, all of which have been instrumental in populating our model with relevant data (Igini).

Hou et al. (167) conducted a study applying the Bass diffusion model to women's fast-fashion shoes over 154 days (172). Their research innovatively modeled pre-sale behaviors—specifically customer attention (how long they looked at shoes) and fitting (how long they tried them on)—measured using RFID sensors (169, 170). The researchers found that combining attention and fitting data allowed for accurate early prediction of sales trends (170). Their Bass diffusion model demonstrated a strong relationship between sales and attention, achieving a very high accuracy ($R^2 \approx 0.999$) (171). This research is relevant to our work as it provides a foundational understanding of how the Bass diffusion model can be applied within the fast fashion context (171). However, while Hou et al. focused solely on predicting sales by modeling adoption rate through attention and fitting, our model expands upon this by integrating the diffusion process into a system dynamics stock-flow structure. In our model, the number of adopters accumulates over time, influenced by factors such as advertising, contact rate, and word-of-mouth. Furthermore, our system dynamics approach allows us to adjust diffusion parameters, like the adoption rate, using EU-specific data, offering a broader applicability.

Another relevant study by Lidia et al. provides multiple causal loop diagrams of the fast fashion industry, using a case study of a firm in Bandung, Indonesia (42–44). Their primary contribution is highlighting and analyzing the fast fashion industry's negative environmental impact, driven by profit motives (42–44). Specifically, Figure 2 in their paper offers a general framework for our system dynamics model by illustrating the link between product attractiveness and increased sales (45). Similarly, Figure 7 demonstrates the interaction of various factors like pricing, worker training, and display rotation with consumer behavior,

influencing the firm's growth and efficiency (49). Overall, Lidia et al.'s model serves as a preliminary guide for our research, offering a rough outline of key causal relationships to consider. However, our model differs by focusing on the EU context and broadening the scope of the impact of certain causal links examined in their firm-specific Indonesian case study.

Furthermore, multiple NGO reports and articles were used to find data for our research. A report by Jeffries and Fall argues that around 7,500 litres of water are used in the production of one pair of jeans, while a report by Oxfam suggests that 2,700 litres of water are required to produce a single T-shirt (Jeffries and Fall; Oxfam). Not only that, but both of these reports suggest policy interventions that are used in our model for consumer awareness; for example, the Oxfam report indicates that consumer awareness can reduce the carbon footprint by 20–30 %, and this number is directly used in our model by inputting a value of 0.25 in the consumer-awareness auxiliary variable (Oxfam). Moreover, a report by the European Environment Agency provides values for GHG emissions, such as the total carbon produced—159 million tonnes—and the carbon produced per person based on clothing, which is 8 kg per year, as well as the total textile carbon emissions, which are 19 kg per year; these figures can then be used to calculate total GHG emissions per item (European Environment Agency). For example, $25.5 \text{ kg CO}_2 \text{ per item} = (159,000,000,000 \text{ kg} \times 8/19) \div (448,000,000 \text{ people} \times 8 \text{ kg} \div 1.3 \text{ kg/item}) \approx 25.5 \text{ kg CO}_2 \text{ per clothing item}$. This value is imputed in our model as well. The idea being that these reports are used to provide data which is then imputed in our system dynamics model.

Valtere et al. and Bailey et al. provide our model with key variables that cause pollution and also offer an idea of multiple intervention policies and their impacts. The main literature gap seems to be a lack of research on fast fashion as a whole and its environmental impact using data specifically from the EU, while there are multiple studies discussing fast fashion at the global level but there seems to be a gap in regards to using EU as a case study. Our study addresses this gap by incorporating and synthesizing data from multiple papers and articles to build a combined system dynamics model, which simulates the environmental impact of fast fashion, as well as how intervention policies can reduce these negative effects by using EU as a case study since there seems to be lack of system dynamics modeling for EU which includes incorporates consumer behaviour.

Methodology:

Research Paradigm: The nature of the fast-fashion industry is a complicated and interconnected process. Many variables may or may not have an effect on each other depending on their nature that cannot be predicted on our own. These interdependencies make it necessary to move beyond a linear cause-and-effect approach and adopt a system dynamics model. The model allows us to provide a comprehensive analysis using a quantitative approach. This is done by incorporating time delays, trends, seasonality, and making use of both reinforcing and balancing loops to identify the underlying drivers within the industry for the benefit of the system.

The fast fashion industry poses an escalating environmental threat due to its dynamics. The industry is characterized by fast moving consumer cycles, low prices and consistent production. There are several key behaviours that occur as a result of the initial infrastructure of the model of the fast fashion industry. Overall, there is an increasing trend with all environmental stocks (GHG Emissions, Waste Generated, Water Consumed, Water Pollution) that are linked towards different variables in the model discussed later.

Exogenous Variables:

Bass Diffusion Model:

- Adoption Fraction: 0.2 ((Okur 368)
- Contact Rate: 20 ((Okur 368)
- Total Population: 4.49e+08 (("Countries in the EU by Population (2025)")
- Advertising Effectiveness: 0.02, Assumption: general meta-analyses of advertising elasticity (Tellis 134-144) suggest that short-term sales response to advertising can be relatively low for frequently purchased goods. Given the nature of fast fashion as a high-volume, trend-driven consumer category, we assume a relatively low initial advertising effect

Policy Intervention Model:

- Bass Awareness Growth: 0.005, Assumption based on natural growth via natural diffusion of information
- Availability of Sustainable Product Options: 0.2, Ranged Benchmark (("Zara Sustainability Goals | How Sustainable is Zara in 2025?")
- Brand Transparency Effort Level: 0.3, Ranged Benchmark ("Zara Sustainability Goals | How Sustainable is Zara in 2025?")
- Circularity Promotion Initiatives: 0.2, ("Zara Sustainability Goals | How Sustainable is Zara in 2025?")

Environmental variables:

- Average Lifespan of an Item: 2 Months, Based on a total of 10 washes
- Average Water Footprint per item: 2000
- BOD per litre of Water Used: 6000
- Average GHG Emission per item: 25.5

Key Variables in the model:

Item Consumption Rate:

The Item Consumption Rate in this model represents the speed at which adopters purchase fast fashion items over time. This rate directly influences the "Cumulative Items Consumed" stock, which in turn drives resource use, pollution, and ultimately waste generation. Technically, this rate is calculated as the product of the number of "Adopters" (a stock representing the current base of fast fashion consumers) and the "Average Consumption

Rate per Adopter" (an auxiliary variable that captures how many items each adopter purchases per unit of time). The "Average Consumption Rate per Adopter" is influenced by a "Trend Speed Multiplier" (reflecting the pace of fashion cycles and thus the need for new items), a "Seasonal Multiplier" (capturing variations in consumption due to seasons or events), and potentially even the level of "Consumer Awareness" (where higher awareness of environmental issues can lead to slightly lower individual consumption). The outflow from the "Cumulative Items Consumed" stock then feeds into the "Item Disposal Rate" based on the "Average Lifespan of the Item." Therefore, the Item Consumption Rate acts as a crucial flow connecting the consumer base to the material throughput of the fast fashion system.

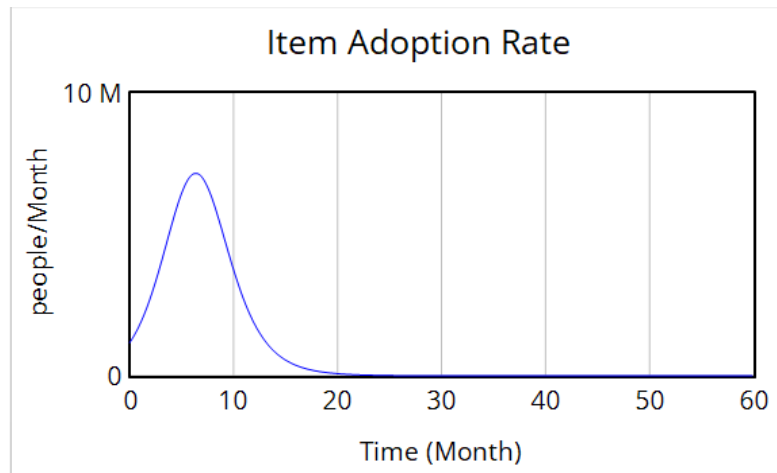
Item Adoption Rate:

The Item Adoption Rate signifies the speed at which individuals transition from being "Potential Adopters" to becoming "Adopters" of fast fashion. This rate determines the growth of the consumer base over time. Technically, this flow is influenced by several factors. "Adoption from Advertising" is calculated as the product of the "Potential Adopters" and an "advertising effectiveness" parameter, scaled by the level of "Consumer Awareness" (as higher awareness makes advertising more or less impactful). "Adoption from Word of Mouth" similarly depends on the number of "Potential Adopters," the current number of "Adopters" (representing the pool of individuals spreading word-of-mouth), and a "contact rate" multiplied by an "adoption fraction" (representing the likelihood of a contact leading to adoption). The "Total Population" also plays a role in defining the pool of "Potential Adopters." The "Item Adoption Rate" is a critical flow that determines the size of the "Adopters" stock, which subsequently drives the "Item Consumption Rate."

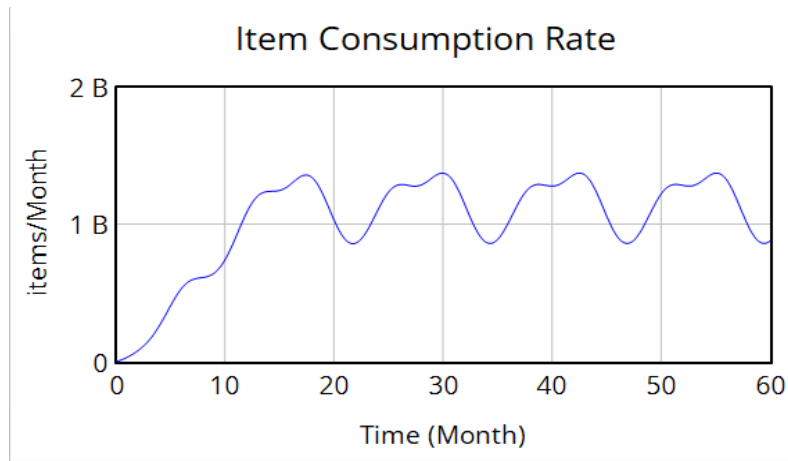
Awareness Generation Rate:

The Awareness Generation Rate represents the speed at which the overall "Consumer Awareness" (a stock in this model) increases or decreases. Technically, this rate is influenced by a multitude of factors, indicating a complex process of information diffusion and attitude formation. A "Base Awareness Growth" provides a background level of increasing awareness. "Awareness Impact on Advertising" and "Awareness Impact on Word of Mouth" suggest that these adoption pathways also contribute to broader awareness levels. Crucially, the "Availability of Sustained Product Options," "Brand Transparency Effort Level" (modulated by "Transparency Effectiveness"), and "Circularity Promotion Initiatives" (modulated by "Circularity Promotion Effectiveness") all contribute positively to awareness, particularly regarding more sustainable aspects. The "Sustainable Product Awareness per Option" scales the impact of sustainable availability on overall awareness. The balance between these positive and negative influences determines the net change in the "Consumer Awareness" stock over time. This stock of "Consumer Awareness" then feeds back to influence both the "Item Adoption Rate" (affecting advertising effectiveness) and potentially the "Average Consumption Rate per Adopter," highlighting the interconnectedness of awareness with consumer behavior.

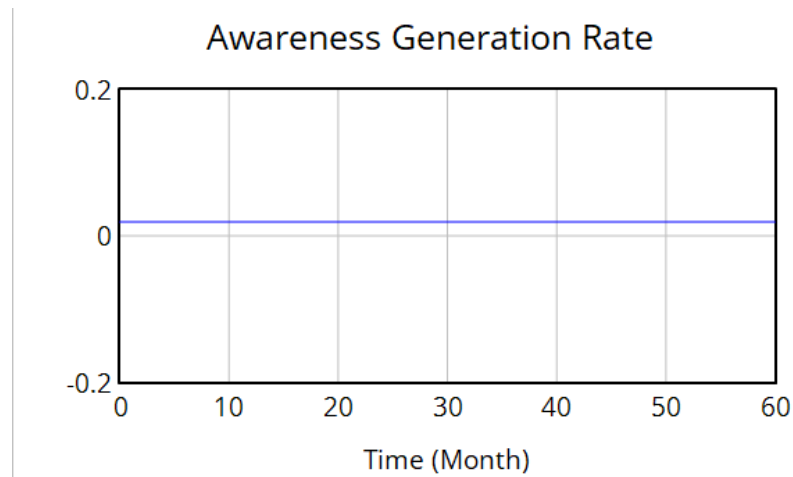
Item Adoption Graph:



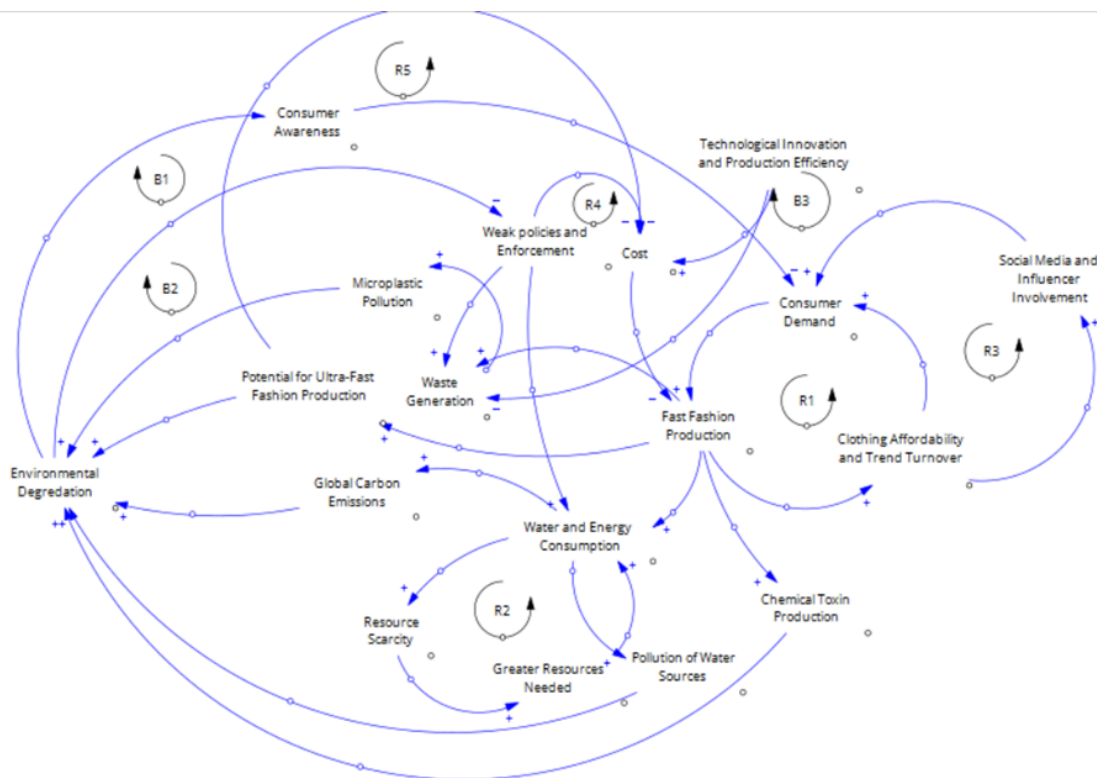
Item Consumption Graph:



Awareness Generation Graph:



Initial causal loop diagram:



The initial CLD depicts the overall fast fashion consumer demand chain and potential balancing and reinforcing loops that could be outlined. Introduction of technological

innovation, efficiency, influencer involvement, and cost have been disregarded during the modeling process to provide a more focused approach on the environmental details

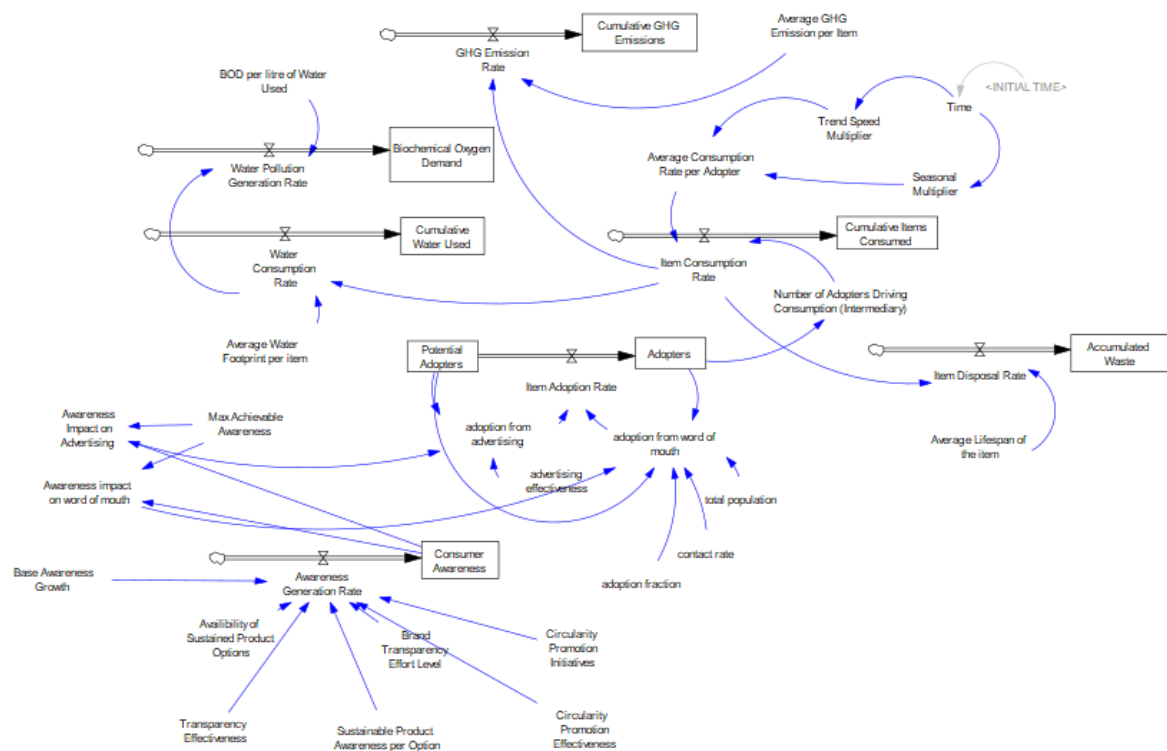
Model Formulation:

Software used: Vensim PLE 10.2.2 version was used for model development.

Timestep: 0.125

Simulation Time: 50 months

Stock and Flow Diagram:



The adoption of consumers leads to the consumption of fast fashion items. This leads to the accumulation of waste, water consumption, and GHG emissions. Consumer awareness acts as a feedback mechanism that modulates adoption rate and subsequently all other environmental degradation stocks.

Equation Formulation

-Awareness Generation Rate: $\text{Base Awareness Growth} + (\text{Availability of Sustained Product Options} * \text{Sustainable Product Awareness per Option}) + (\text{Brand Transparency Effort Level} * \text{Transparency Effectiveness}) + (\text{Circularity Promotion Initiatives} * \text{Circularity Promotion Effectiveness})$

-Item Adoption Rate: $\text{adoption from advertising} + \text{adoption from word of mouth}$

-Adoption from word of mouth: $\text{contact rate} * \text{adoption fraction} * \text{Potential Adopters} * (1 - \text{Awareness impact on word of mouth}) * (\text{Adopters} / \text{total population})$

-Awareness Impact on advertising: $\text{Consumer Awareness} / \text{Max Achievable Awareness}$

-Awareness Impact on word of mouth: $\text{Consumer Awareness} / \text{Max Achievable Awareness}$

-Adoption from advertising: $\text{advertising effectiveness} * \text{Potential Adopters} * (1 - \text{Awareness Impact on Advertising})$

-Water Consumption Rate: $\text{Item Consumption Rate} * \text{Average Water Footprint per item}$

-Water Pollution Generation Rate: $\text{BOD per litre of Water Used} * \text{Water Consumption Rate}$

-Item Disposal Rate: $\text{DELAY1}(\text{Item Consumption Rate}, \text{Average Lifespan of the item})$

-Average Consumption Rate per Adopter: $20 * \text{Trend Speed Multiplier} * \text{Seasonal Multiplier}$

-Trend Speed Multiplier: $1 + (\text{SIN}(\text{Time}/2)) * 0.2$

-Seasonal Multiplier: $1 + (\text{SIN}(\text{Time} + 2) * 2 * 3.14 / 12) * 0.15$

-Item Consumption Rate: $\text{Average Consumption Rate per Adopter} * \text{Number of Adopters Driving Consumption (Intermediary)}$

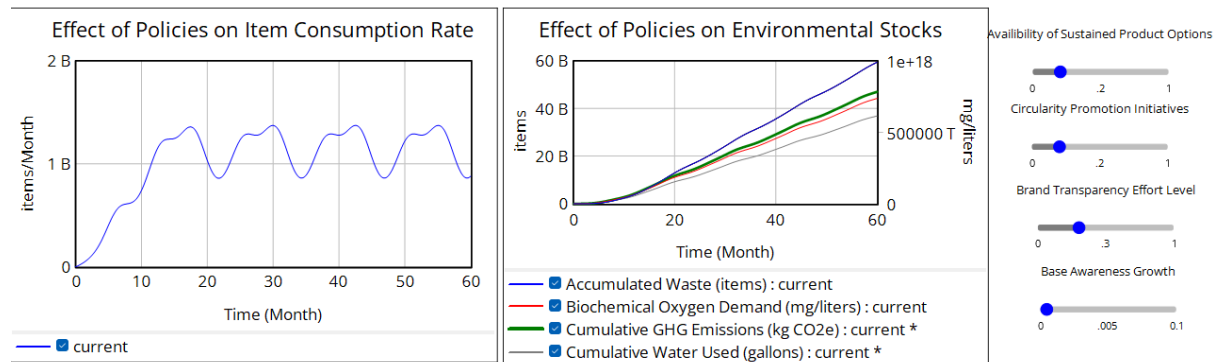
-GHG Emission Rate: $\text{Average GHG Emission per Item} * \text{Item Consumption Rate}$

Results: Policy Analysis and Scenario Development

Baseline Case:

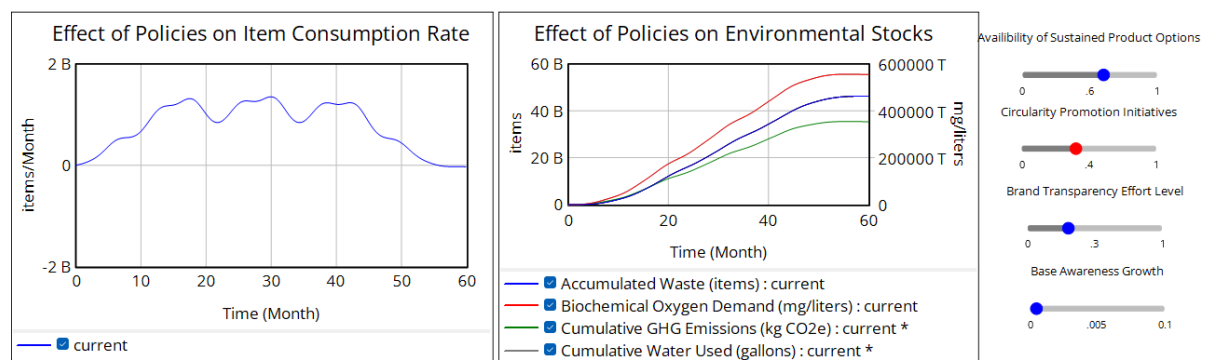
An initial level was set for all three policies in accordance with the current figures released by fast fashion brands. 20% of the total products produced were said to be sustainably made. 20% of Zara's marketing budget was dedicated towards initiatives towards the promotion of a circular economy. A brand transparency index was used to gauge the current level by Zara and their efforts that would improve consumer awareness. The base awareness growth was set at 5% to mimic average awareness of sustainability for a major demand market. The current item consumption rate depicts-with trends and seasonality-a consumption of 1.3

billion items per month with an increasing trend in the first 12 months depicting the enormous diffusion spread of the industry. The accumulated waste simulated 58 billion items consumed by the end of 5 years. Alongside cumulated GHG emissions of 1.5 trillion kgCO₂e, and 121 trillion gallons of water consumed every 5 years. It is also important to note that the environmental stock graphs are on an increasing trend that will exceed the current maximum values as the simulation time is increased.

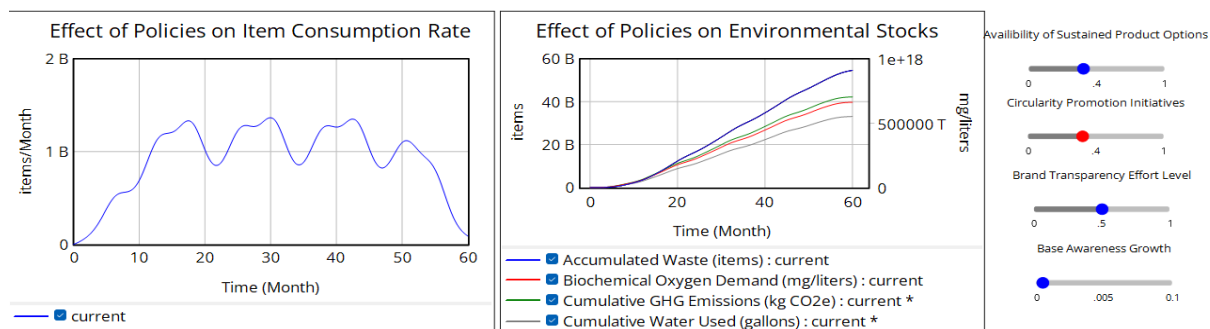


Policy Interventions

At a 60% level of sustainable product options, 40% of circularity promotion initiatives, and a 30% brand transparency level, item consumption rate reaches 0 by the end of 5 years depicting a complete shift from fast fashion brands that are unsustainable. This is also translated into the environmental stocks. The value for accumulated waste falls by 23 billion items, while the GHG emissions reduce to 1.1 trillion kgCO₂e at their maximum values. Introduction of these interventions have also broken the increasing trend of the environmental stocks and limited them to values below the previous maximum.



However, a more realistic depiction that may be translated in the real world is the availability of 40% of sustainable product options for consumers with circularity initiatives set at 40% as well. Brand transparency should be considered an important aspect for business especially in fast fashion and is set at 50% to initiate change. It is also important to note that the current base awareness growth for all scenarios is set at 5% and that may see an increasing trend in the future as newer generations are caring more and more about the environment.



The current settings allow for the consumption rate to have a steady decrease after the first few years and reduce the accumulated waste to 54 billion items, GHG Emissions to 1.4 kgCO₂e, and total water used to 110 trillion gallons.

Conclusion:

While the current model offers valuable insights into the fast fashion industry's environmental impact, it has several limitations. First, the simulation is restricted to the European Union, limiting the model's applicability to regions with different consumption patterns, regulatory frameworks, and waste management systems. Second, the model assumes a uniform water footprint across all clothing items, despite significant variation between different products—for example, jeans and T-shirts have vastly different virtual water requirements. Additionally, the model simplifies consumer behavior and omits important psychological and cultural drivers that influence fast fashion demand.

Future extensions could address these gaps by incorporating global data to simulate fast fashion dynamics across multiple continents, enabling comparative analysis between the Global North and South. The model could also be enhanced by integrating the production side of the supply chain, including emissions from transportation, the role of renewable energy, and other factors necessary for a more comprehensive life cycle assessment. Furthermore, incorporating behavioral feedback loops, such as the influence of education campaigns or social media on sustainable purchasing habits, could improve the model's ability to simulate long-term shifts in consumer behavior. Finally, introducing economic and stakeholder variables—such as brand incentives, policy enforcement, and price elasticity—would make the model more robust and relevant for real-world policy design.

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